Design Note DN-7: Plate Heat Exchanger for the District Heating Loop

1 Duty

The exchanger transfers the heat of the primary loop to the secondary loop of the Riverside district. The design duty is 2.4 megawatts, with the primary water entering at 95 degrees Celsius.

2 Sizing method

2.1 Logarithmic mean temperature difference

The driving temperature difference is the logarithmic mean of the differences at both ends:

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}$$

With 25 kelvin at the hot end and 15 kelvin at the cold end, the logarithmic mean is 19.6 kelvin.

2.2 Required area

The heat transfer area follows from the duty, the overall heat transfer coefficient U and the mean temperature difference:

$$A = \frac{Q}{U \, \Delta T_{lm}}$$

With U equal to 4,500 watts per square metre and kelvin, the required area is 27.2 square metres; the selected unit has 31 square metres, which leaves a margin for fouling.

3 Materials

The plates are made of stainless steel 316L, 0.5 millimetres thick, and the gaskets of EPDM rated for 130 degrees Celsius.